Airbnb, 2018: similar homes are nearest neighbours

This weekend: Friday 08:00, the video store

Each home is a list of 32 numbers, learned from over 800 million search sessions. Similar homes: its 12 nearest neighbours.

Result, A/B test

+21% *click rate of the row of similar homes*

+4.9% *guests who found their booking here*

What it cost

A new home has no sessions yet: its numbers are the average of 3 similar homes nearby.

What Airbnb decided

Learn from the homes guests booked as well as clicked; launch after an online test.

Grbovic and others (Airbnb), Airbnb Tech Blog, 13 March 2018. Grbovic and Cheng, KDD 2018.

Netflix, 2009: the prize went to factorization

This weekend: Friday 10:00, the matchmaker

Netflix shared 100 million ratings and offered $1 million for guesses 10% better than its own system.

Result, on hidden ratings

10.06% *better than Netflix's own system*

100+ *models by 2008, most factorizations*

What it cost

Researchers matched public IMDb ratings to the anonymous data: 2 of about 50 IMDb users identified.

What Netflix decided

After FTC privacy concerns and a lawsuit, cancel Netflix Prize 2 (March 2010).

Netflix Prize rules (2006) and results (2009). Koren, Bell, Volinsky, IEEE Computer 2009. Narayanan and Shmatikov, IEEE S&P 2008. Netflix blog and FTC letter, 12 March 2010.

Adyen, 2025: a better bandit approves more payments

This weekend: Friday 12:00, the bandits

Most payments can reach the bank in 2 to 15 ways. Epsilon-greedy picks the best bet, and now and then another.

Result, A/B test

+0.1% *approval rate, smarter exploring*

9M *more approvals a year, estimated*

What it cost

Exploring risks declines. Smarter exploring left fewer declines in the data: the next model did 0.2% worse.

What Adyen decided

Rules first: contracts and card network rules remove options before the bandit picks.

Vangara and Egg (Adyen), OARS workshop, KDD 2025. van Rooijen (Adyen), Adyen Knowledge Hub, 2 November 2020.

Netflix, 2012: the winning blend did not ship

This weekend: Friday 14:00, gradient descent

Netflix put the two strongest prize models into production: a matrix factorization and a neural network.

Result, on hidden ratings

8.43% *107 models, 2007*

10.06% *hundreds of models, 2009*

What it cost

Built for 100 million ratings, the models had to learn from more than 5 billion, and keep learning as members rated.

What Netflix decided

Ship those two. Newer methods: the offline gains did not seem worth the engineering.

Amatriain and Basilico (Netflix), Netflix Tech Blog, 6 April and 20 June 2012. Funk, blog, 11 December 2006.

Which technique answers which question

The business question	*The technique*	*The cases*
Which products, or clients, are like this one?	Nearest neighbours	*Airbnb, 08:00* *also Amazon, Pinterest*
Which unseen product will this client want?	Matrix factorization	*Netflix, 10:00, 14:00* *also Spotify*
Which option to use, while still learning which works?	Bandits	*Adyen, 12:00* *also Yahoo, Microsoft, Spotify, Amazon*

Amazon, IEEE Internet Computing 2003. Pinterest, KDD 2018. Spotify, slides 2014 and 2015. Yahoo, WWW 2010. Microsoft, arXiv 2016. Spotify, RecSys 2018. Amazon, KDD 2017.